Technology Stack

Date	15 July 2026
Team ID	xxxxxx
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	**Frontend / Client-Side** *(e.g., React, Angular, HTML/CSS)*	HTML, CSS, JavaScript, Bootstrap	Responsive UI

2	**Backend / Server-Side** *(e.g., Node.js, Python/Django, Java)*	Python (Flask)	Backend and AI integration
3	**Database / Data Storage** *(e.g., MySQL, MongoDB, PostgreSQL)*	Firebase Firestore	Realtime cloud database
4	**Cloud / Hosting / Deployment** *(e.g., AWS, Azure, Heroku, Docker)*	Google Cloud	Scalable deployment
5	**Version Control & CI/CD**	GitHub	Version control and collaboration

	(e.g., GitHub, GitLab, Jenkins)		
6	**Third-Party APIs / Other Tools** *(e.g., Stripe, Twilio, Google Maps)*	Google Gemini API, OpenCV	Emotion detection and AI support